For each of the following polyhedra:

1. represent it graphically,
2. identify the corresponding polyhedron in standard form,
3. enumerate the feasible basic solutions, and identify the corresponding vertex on the polyhedron, and,
4. for each feasible basic solution, mention if it is degenerate or not.

$$P = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 | x_1 + x_2 \leq 1, -x_1 + 2x_2 \leq 2, x_1 \geq 0, x_2 \geq 0 \right\},$$

$$Q = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 | x_1 + x_2 \leq 1, x_1 - x_2 \leq 1, x_1 \geq 0, x_2 \geq 0 \right\}.$$